November 14, 2022

APS110 – Lecture 25

See hand written notes for drawings.

Electromagnetic spectrum

- How does light interact with materials? → optical transparency
- Recall: plexiglass has an amorphous structure and so light can pass right through it and so it is optically transparent
- But, this is <u>not</u> the whole story → why can other amorphous materials be opaque?
  - o Interactions of the material with photons of energy affect the optical transparency
  - o Obviously there can't be any scattering events occurring in the material if we want it to be transparent

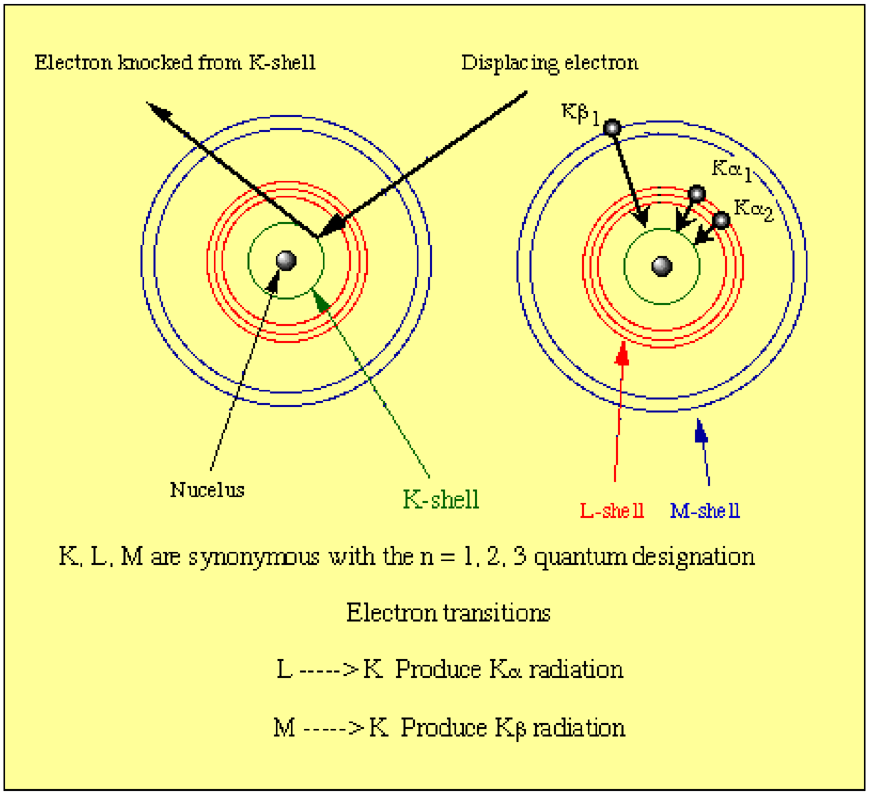
Electronic structure

- **Bohr model** → describes a central nucleus with orbitals of electrons surrounding it
  - o n = 1 → first electron shell, closest to nucleus
  - o Electrons can <u>only</u> exist in these defined shells → the electron energy level is **quantized**
    - ▪ Electrons can jump whole energy levels, ex. from n=1 to n=2, but not incremental energy levels (ex. there is no n=1.5)
- When a photon of energy interacts with this atom, if it has sufficient energy it could promote an electron to a higher energy state
  - o When that electron returns to its original energy state, drops back down, it will emit a photon
  - o This photon released will have a defined energy

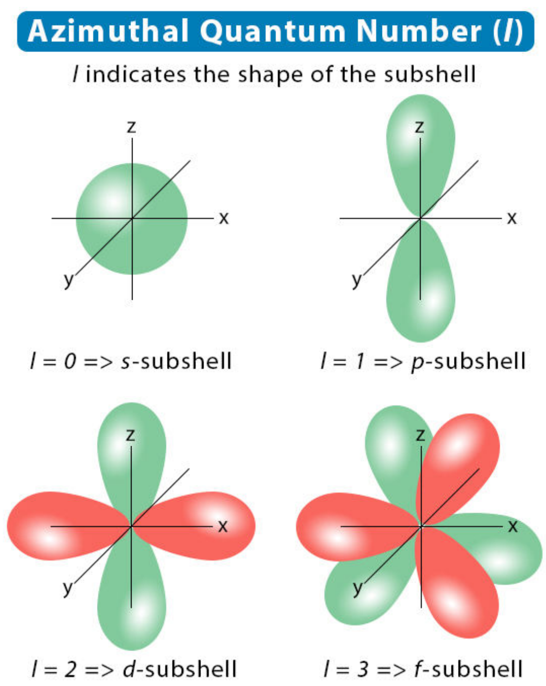
- As we get deeper into the discussion of electron energy levels, we learn that there is actually a bit more fine spacing to the whole number quantization (**principle quantum number**)
  - o There are a few more finely spaced energy levels, each close to n=1,2,3,4, etc.
- We describe these energy levels by **quantum numbers**
  - o There are 4 different numbers in the **wave-mechanical model**

Wave-Mechanical Model

1. **Principle Quantum Number** (QN) → describes the size of the electron probability distribution (the size of the shell that the electron could exist in)
   - n = 1,2,3,4…
   - Sometimes these are described with letters: K, L, M, N
     - o These are synonymous with the n = 1,2,3,4…
   - Ex. Cu K-α
     - o α tells us it is a drop of energy of one step (one principle QN)



2. **Angular Momentum QN** → describes the **shape**
   - l = 0,1,2,3,...(n-1)
   - More often, we use lower case letters **s,p,d,f** top describe the angular momentum QNs



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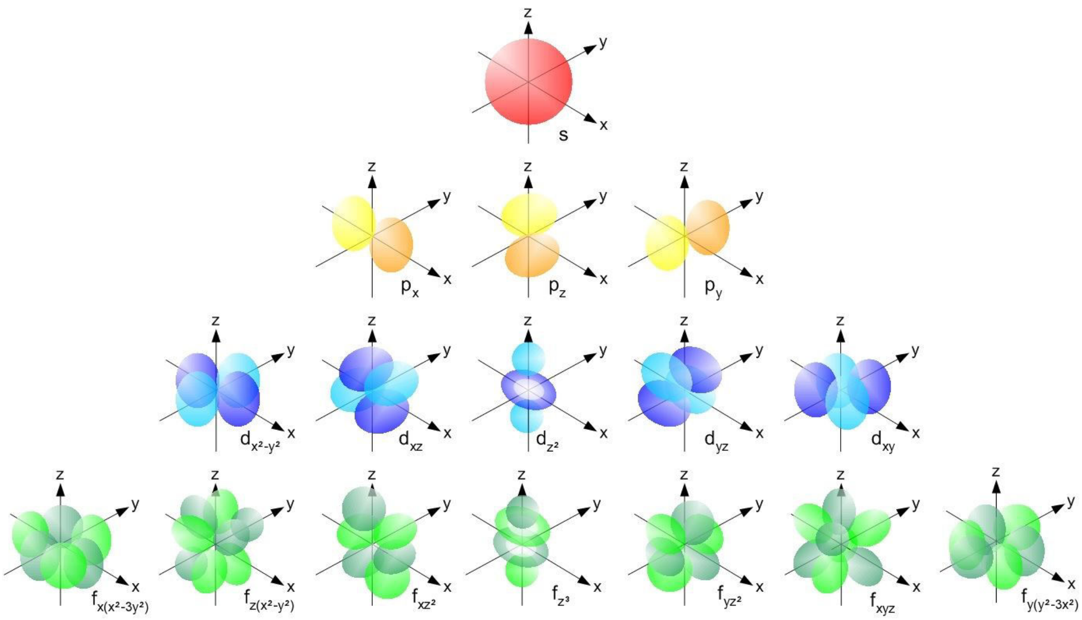
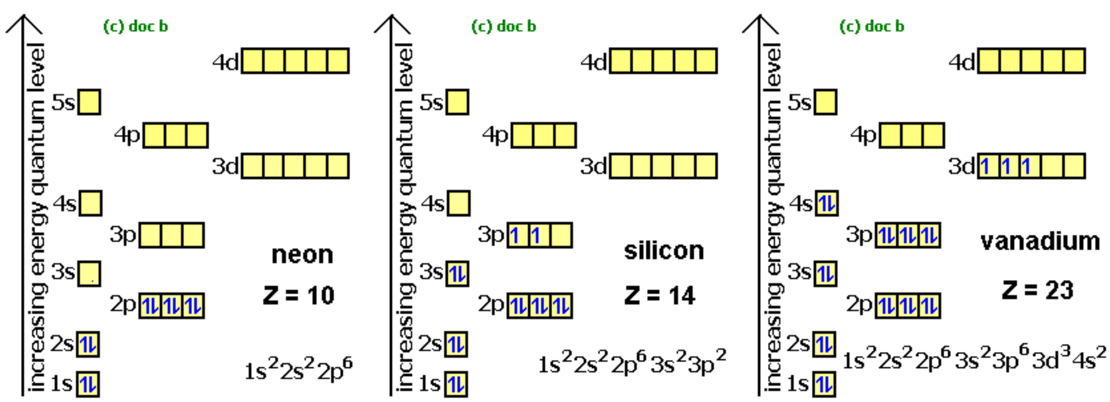
3. Magnetic QN

- $-l \le m_l \le l$

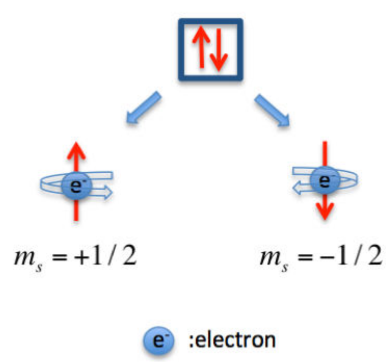


- Electron box diagrams:



4. **Spin QN** → "spin"

- $m_s = +/- ½$



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Now, how are these energy levels positioned relative to each other?

- We categorize energy level by how much energy is required to ionize electrons in that energy level
- The electrons in the levels closest to the nucleus are the **hardest** to ionize → therefore they have the **<u>lowest</u>** energy level